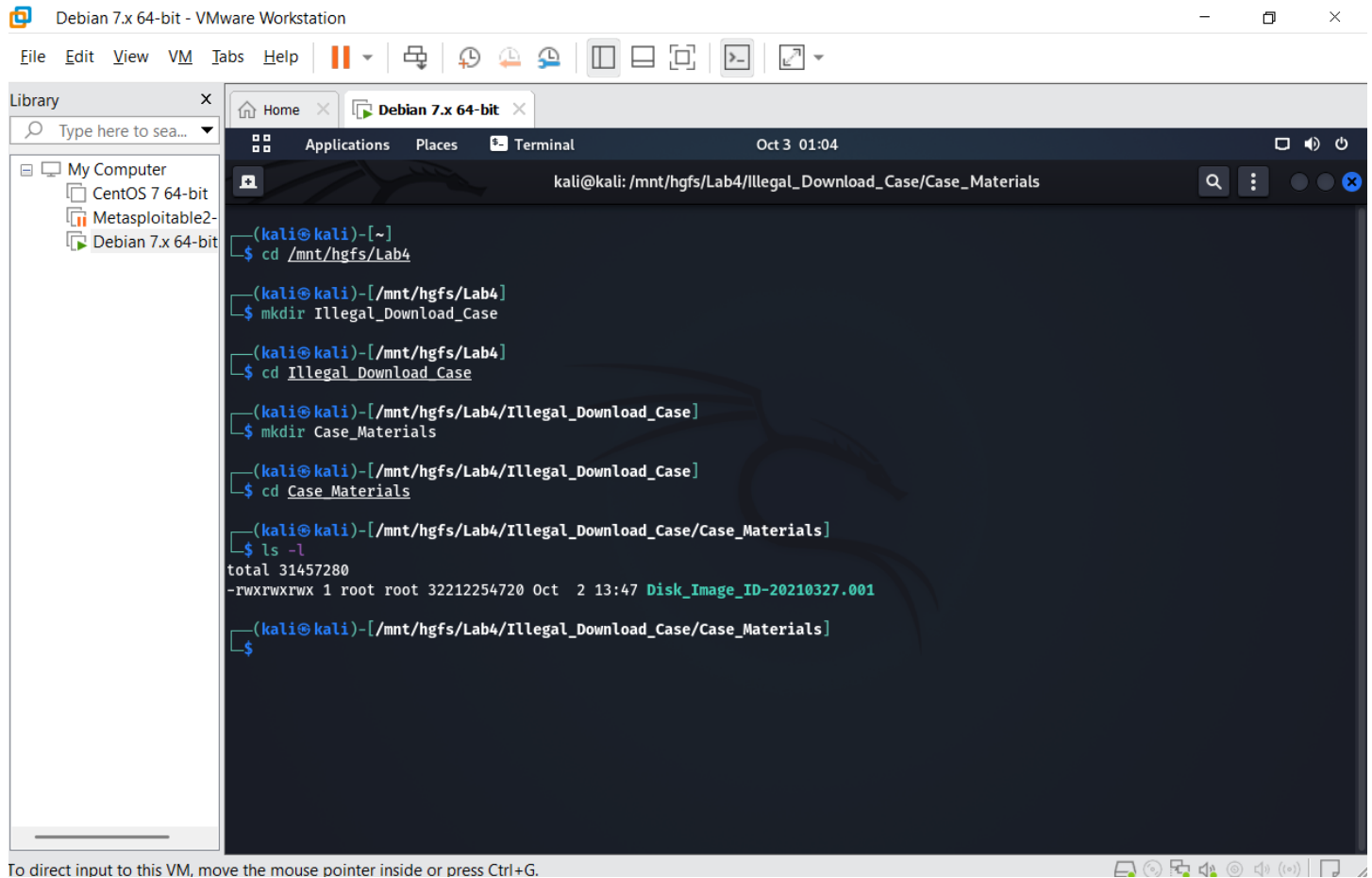
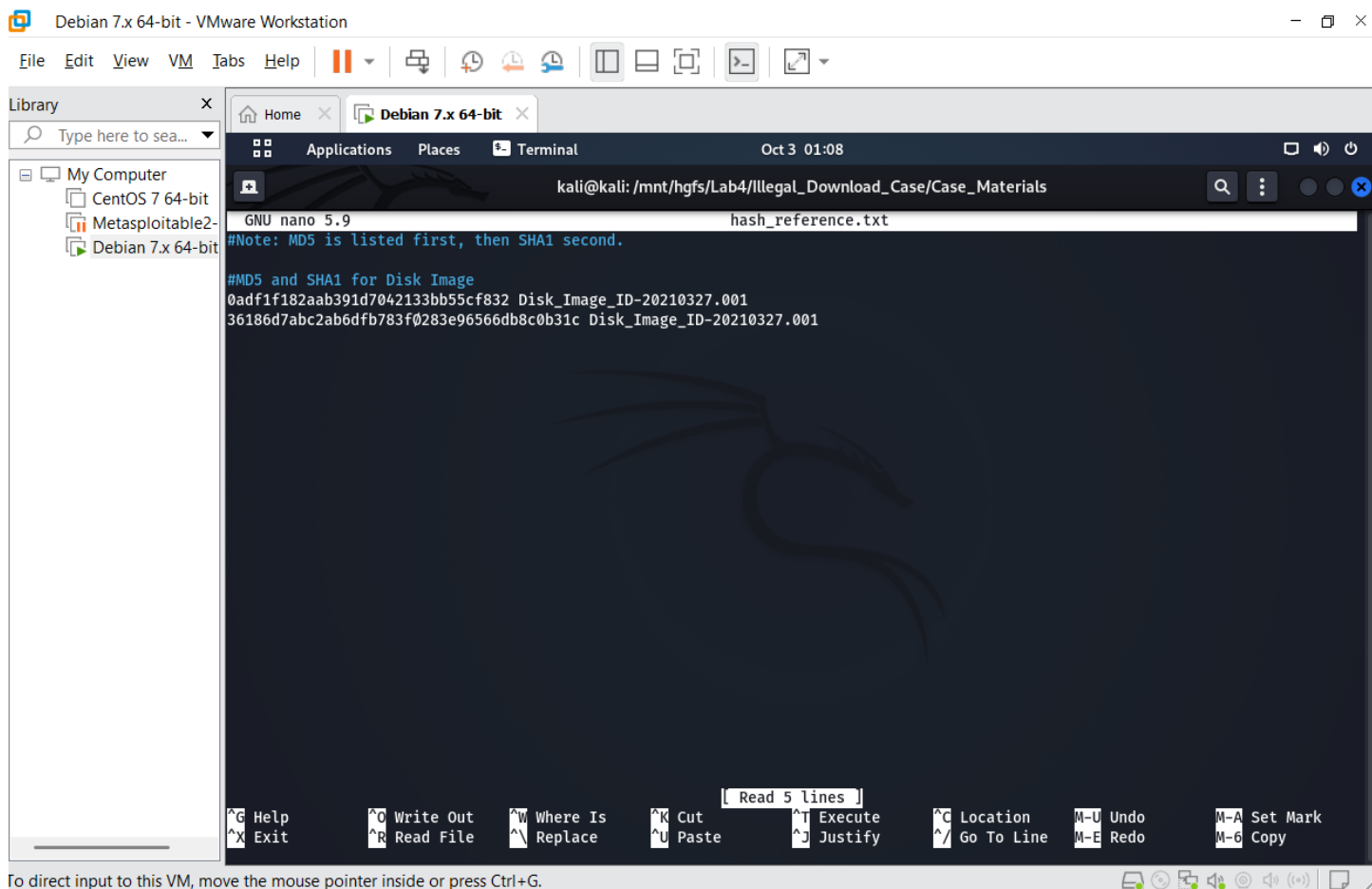


LAB 4: Disk Image and Partitions

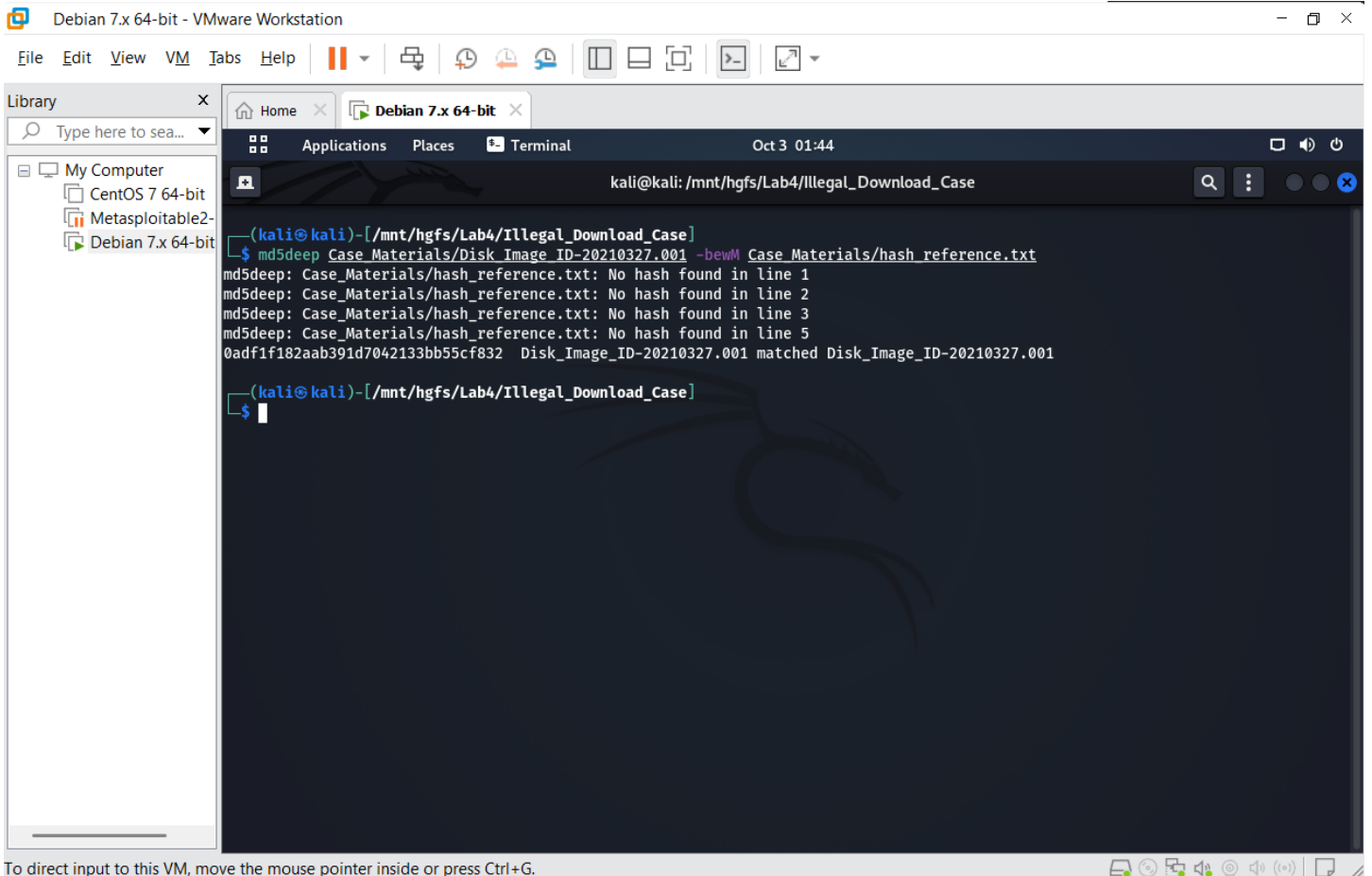
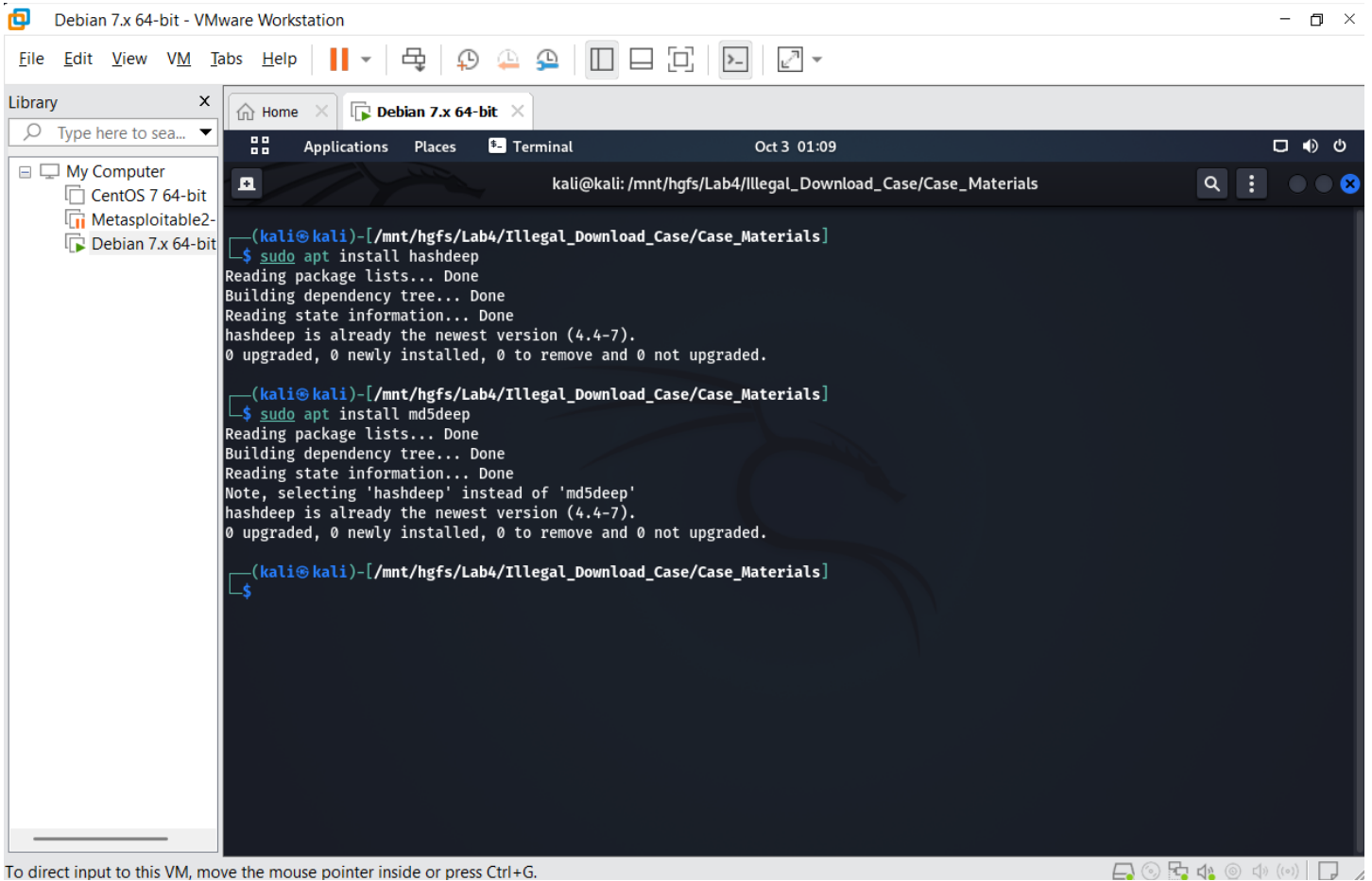
1. Verify the integrity of the disk image

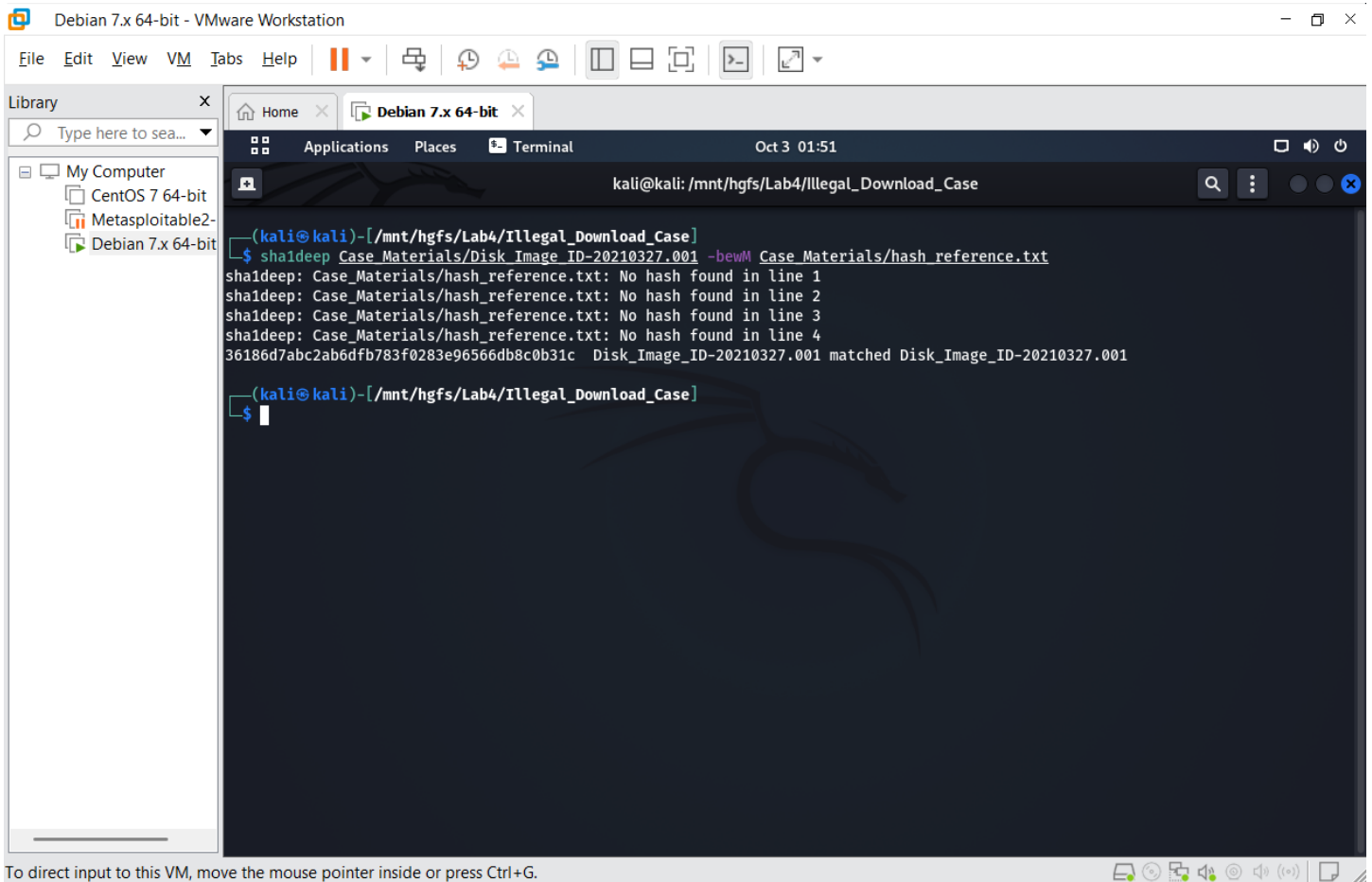


ảnh: tải file Disk Image cần thiết cho bài lab



ảnh: file hash_reference.txt ghi lại mã hash gốc (original hash) được cung cấp của file Disk_Image_ID-20210327.001

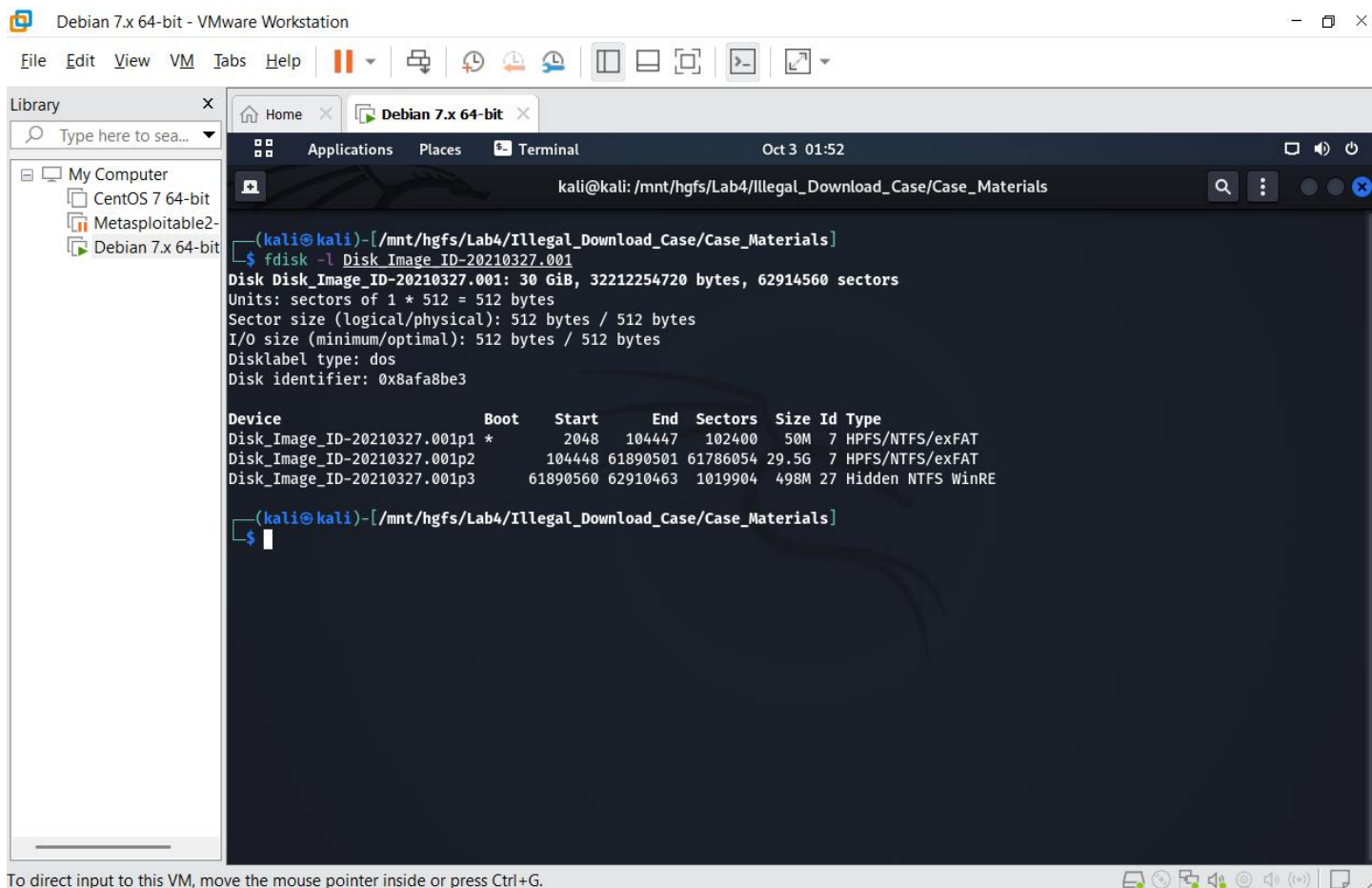




2 ảnh trên: cài đặt công cụ hashdeep. Sử dụng công cụ để tính mã hash MD5, SHA của file Image đã tải và so sánh với mã hash đã được cung cấp trước đó (original hash) trong file hash_reference.txt, nếu trùng khớp thì file đã được download về nguyên vẹn

2. Identify the OS of the system as well as its name, accounts, and partitions.

Mục tiêu: xác định có bao nhiêu phân vùng trong file Image, bắt đầu ở đâu.



ảnh: dùng lệnh `fdisk -l` để xác định offset của từng phân vùng và ghi lại để dùng cho bước tiếp theo

Debian 7.x 64-bit - VMware Workstation

File Edit View VM Tabs Help

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- Metasploitable2
- Debian 7.x 64-bit

Applications Places Terminal

Oct 3 01:54

kali@kali: /mnt/hgfs/Lab4/Illegal_Download_Case/Case_Materials

-k password: Decryption password for encrypted volumes

(kali@kali)-[/mnt/hgfs/Lab4/Illegal_Download_Case/Case_Materials]

```
$ fsstat -o 2048 Disk Image ID-20210327.001
```

FILE SYSTEM INFORMATION

File System Type: NTFS
Volume Serial Number: 18EC42BBEC4292C4
OEM Name: NTFS
Volume Name: System Reserved
Version: Windows XP

METADATA INFORMATION

First Cluster of MFT: 4266
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 256
Root Directory: 5

CONTENT INFORMATION

Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 12798
Total Sector Range: 0 - 102398

\$AttrDef Attribute Values:
\$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
\$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
\$FILE_NAME (16) Size: 60-520 Flags: Resident Today

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

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Oct 3 01:55

kali@kali: /mnt/hgfs/Lab4/Illegal_Download_Case/Case_Materials

\$EA (224) Size: 0-65536 Flags:
\$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident

(kali@kali)-[/mnt/hgfs/Lab4/Illegal_Download_Case/Case_Materials]

```
$ fsstat -o 104448 Disk Image ID-20210327.001
```

FILE SYSTEM INFORMATION

File System Type: NTFS
Volume Serial Number: E8DE4350DE4315EA
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION

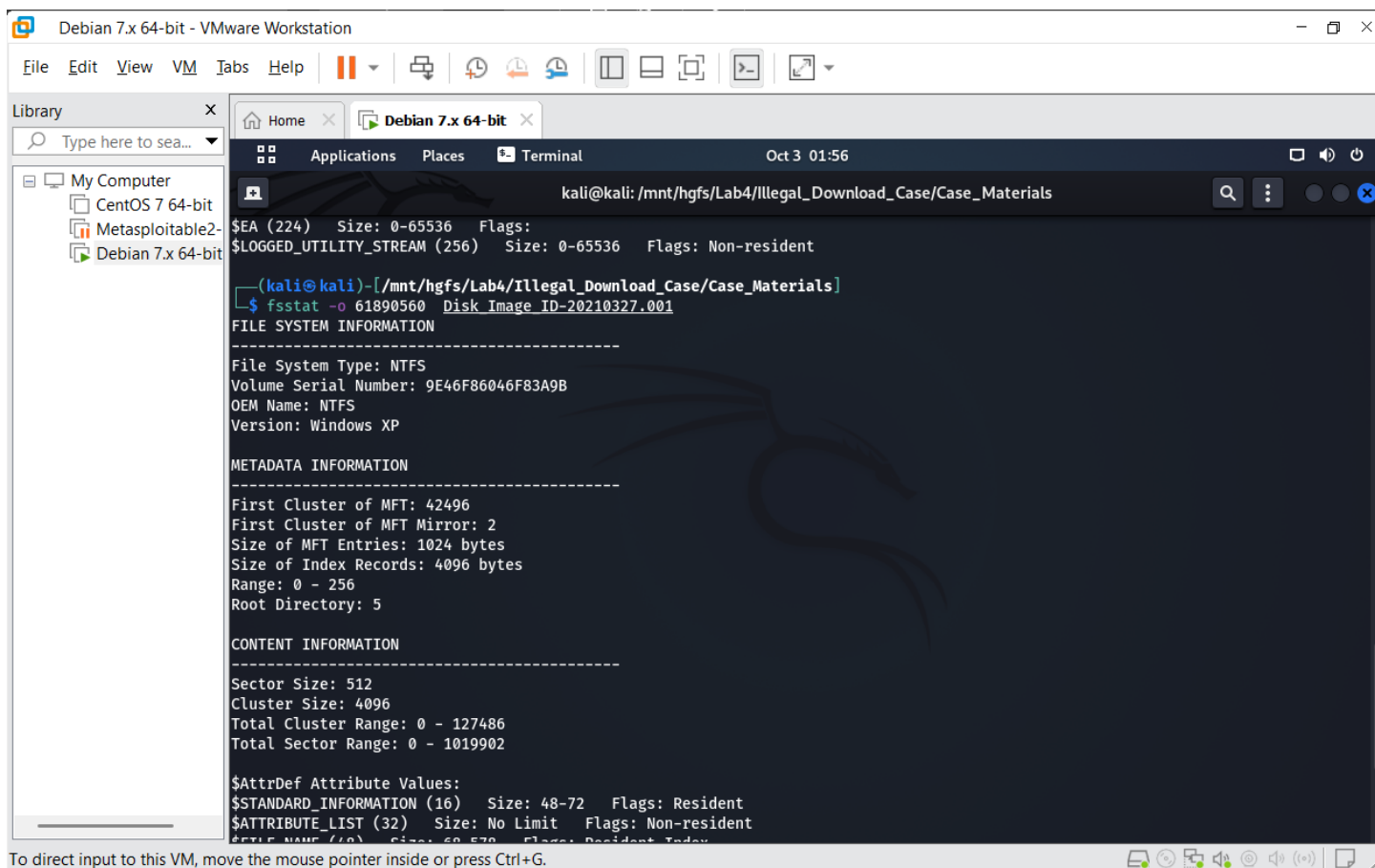
First Cluster of MFT: 786432
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 226304
Root Directory: 5

CONTENT INFORMATION

Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 7723255
Total Sector Range: 0 - 61786052

\$AttrDef Attribute Values:
\$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
\$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
\$FILE_NAME (16) Size: 60-520 Flags: Resident Today

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
kali@kali: /mnt/hgfs/Lab4/Illegal_Download_Case/Case_Materials
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident

(kali@kali)-[/mnt/hgfs/Lab4/Illegal_Download_Case/Case_Materials]
$ fsstat -o 61890560 Disk_Image_ID-20210327.001
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: 9E46F86046F83A9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 42496
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 256
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 127486
Total Sector Range: 0 - 1019902

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (16) Size: 60-570 Flags: Resident Today
```

3 ảnh trên: dùng fsstat xác định loại hệ thống tập tin (NTFS, EXT4, FAT32...), serial number, cấu trúc của từng phân vùng

ta thấy:

Partition 1: File System Type: NTFS

Volume Serial Number: 18EC42BBEC4292C4

Partition 2: File System Type: NTFS

Volume Serial Number: E8DE4350DE4315EA

Partition 3: File System Type: NTFS

Volume Serial Number: 9E46F86046F83A9B

Partition Table			MS-DOS				
Partition	Flag	Start	End	Sectors	Size	File System	Serial #
1 st Partition – System Reserved	Boot	2048	104447	102400	50 MB	NTFS	18EC42BBEC 4292C4
2 nd Partition	-	104448	61890501	61786054	29.5 GB	NTFS	E8DE4350DE 4315EA
3 rd Partition	-	61890560	62910463	1019904	498 MB	NTFS/Hidden NTFS WinRe	9E46F86046 F83A9B

Giải thích ý nghĩa tham số:

- **Partition:** số phân vùng (các “ngăn” trong ổ đĩa).
- **Offset (sector):** vị trí bắt đầu của phân vùng.
- **File System:** hệ thống tập tin, quyết định cách dữ liệu được tổ chức.
- **Serial Number:** định danh duy nhất cho volume, dùng để xác thực và truy vết.
- **Size:** Dung lượng của phân vùng.